

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

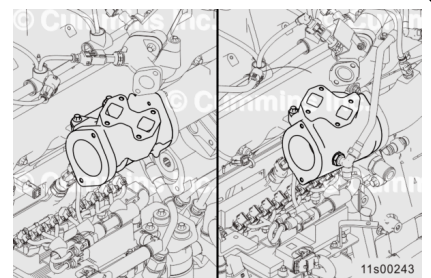
- No recommended Cummins® service tools required

Additional Service Items

- Mild abrasive pad.

General Information

The Exhaust Gas Recirculation (EGR) mixer has two configurations, with and without water jacket.



Preparatory Steps

⚠ WARNING ⚠

If natural gas leaks are present, do not store the equipment inside or in any area that is covered. Severe personal injury can result from asphyxiation or explosion.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Natural gas is explosive and flammable. Always be sure to maintain adequate ventilation in the work area. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas with shared ventilation to reduce the possibility of severe personal injury or death when working on a natural gas system.

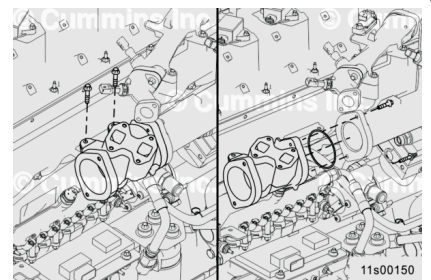
- Before removing any component, close the vehicle main fuel supply valve.
- Start the engine and allow it to operate at idle. Allow the engine to operate until it shuts down.
- Disconnect the batteries. See equipment manufacturer service information.
- Remove the air intake connection. Refer to Procedure 010-080 in Section 10.
(/qs3/pubsys2/xml/en/procedures/1016/1016-010-080.html)
- Remove the air intake connection adapter. Refer to Procedure 010-131 in Section 10.
(/qs3/pubsys2/xml/en/procedures/1016/1016-010-131.html)
- Remove the throttle body assembly. Refer to Procedure 005-052 in Section 5.
(/qs3/pubsys2/xml/en/procedures/1016/1016-005-052-tr.html)
- Remove the air/fuel mixer. Refer to Procedure 005-003 in Section 5.
(/qs3/pubsys2/xml/en/procedures/1016/1016-005-003.html)
- Remove the EGR valve. Refer to Procedure 011-022 in Section 11.
(/qs3/pubsys2/xml/en/procedures/1016/1016-011-022.html)
- Remove the water jacket line if equipped. Refer to Procedure 008-152 in Section 8.
(/qs3/pubsys2/xml/en/procedures/1016/1016-008-152.html)

Remove

Remove the two capscrews holding the EGR mixer to the air intake manifold.

Remove the two capscrews holding the EGR mixer to the air intake connection.

Remove the EGR mixer.



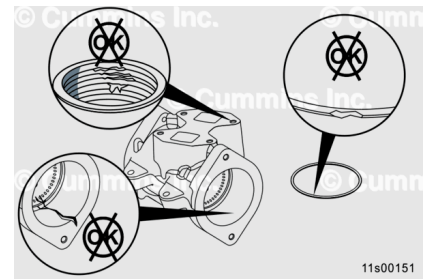
Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Inspect the EGR mixer for dirt, debris, or damage.

Inspect and replace the o-rings if damage is found.

Inspect the passages for plugging.

If the mixer passages are found to be obstructed, the following cleaning steps shall be taken:

- Use a mild abrasive pad and water to gently scrub the material from the surface of the mixer.

Note : Do not use sandpaper or other materials that can leave particulate contamination in the intake.

- Thoroughly rinse the EGR mixer in clean water to remove any debris.

Note : If the contamination can not be removed from the fuel ports, or if the ports are damaged during cleaning, the mixer must be replaced.

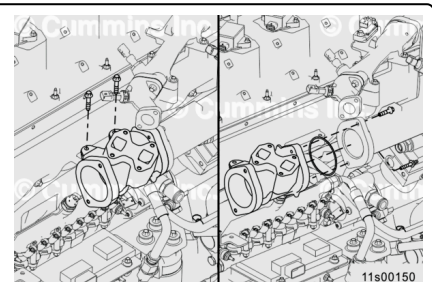
Install

Install the two capscrews holding the EGR mixer to the air intake manifold.

Install the two capscrews holding the EGR mixer to the air intake connection.

Tighten the capscrews.

Torque Value: 18 n•m [159 in-lb]



Finishing Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the water jacket line if removed. Refer to Procedure 008-152 in Section 8.
(/qs3/pubsys2/xml/en/procedures/1016/1016-008-152.html)
- Install the EGR valve. Refer to Procedure 011-022 in Section 11.
(/qs3/pubsys2/xml/en/procedures/1016/1016-011-022.html)
- Install the air/fuel mixer. Refer to Procedure 005-003 in Section 5.
(/qs3/pubsys2/xml/en/procedures/1016/1016-005-003.html)
- Install the throttle body assembly. Refer to Procedure 005-052 in Section 5.
(/qs3/pubsys2/xml/en/procedures/1016/1016-005-052-tr.html)
- Install the air intake connection adapter. Refer to Procedure 010-131 in Section 10.
(/qs3/pubsys2/xml/en/procedures/1016/1016-010-131.html)
- Install the air intake connection. Refer to Procedure 010-080 in Section 10.
(/qs3/pubsys2/xml/en/procedures/1016/1016-010-080.html)
- Connect the batteries. See equipment manufacturer service information.
- Slowly open the vehicle main fuel supply valve.
- Operate the engine and check for leaks.